TUGAS 1 PRAKTIKUM ANALISIS REGRESI

Nama: Muhammad Daffa Fadhilah Athif
NIM: M0401241085

A. Parameter Regresi

Model Regresi linear sederhana:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1,2,\ldots,30 \text{ (karena 30 amatan)}$$

Keterangan:

$Y_i = hargarumah\ (rupiah)$

$X_i = LuasTanah\ (m^2)$

$\beta_0 = intersep$

$\beta_1 = Koefisien\ Regresi$

$\varepsilon_i\ =\ Galat$

1. Koefisien Regresi ($\widehat{\beta_1}$)

$$\widehat{\beta_1} = \frac{\Sigma(y_i - \bar{y})(x_i - \bar{x})}{\Sigma(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

$$\widehat{\beta_1} = \frac{2{,}71518E+12}{371337{,}5}$$

$$\widehat{\beta_1} = 7311897$$

Setiap penambahan 1 m² luas tanah, harga rumah rata-rata meningkat sebesar Rp. **7.311.897**, dengan asumsi faktor lain tetap.

2. Intersep ($\widehat{\beta_0}$)

$$\widehat{\beta_0} = \bar{y} - \widehat{\beta_1}\bar{x}$$

$$\widehat{\beta_0} = 1762300000 - (7311897)(123{,}5333)$$

$$\widehat{\beta_0} = 859037026{,}70$$

Jika luas tanah **0 m²**, maka harga rumah yang diprediksi adalah sekitar **Rp 859.037.027**.

3. Model Regresi

$$\hat{Y} = 859037027 + 7311897X$$

Koefisien regresi menunjukkan bahwa setiap kenaikan **1 m² luas tanah** akan meningkatkan harga rumah rata-rata sebesar **Rp 7.311.897**.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  859037027  239339148   3.589  0.00125 **
land_size_m2   7311897    1439651   5.079 2.23e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

B. Uji Hipotesis dan Selang Kepercayaan

1. Uji hipotesis dan Selang kepercayaan (1- α) bagi parameter regresi $\beta 1$

$H_0: \beta_1 = 0$

$H_1: \beta_1 \neq 0$

Hipotesis nol menyatakan bahwa luas tanah **tidak berpengaruh** terhadap harga rumah, sedangkan hipotesis alternatif menyatakan bahwa luas tanah **berpengaruh** terhadap harga rumah.

Rumus Statistik Uji:

$$t_{\text{hitung}} = \frac{\widehat{\beta_1}}{S\widehat{\beta_1}}$$

Dengan:

$$S\widehat{\beta_1} = \sqrt{\frac{JKS}{n-2} / \Sigma(x_i - \bar{x})^2}$$

$$JKS = \Sigma(y_i - \hat{y})^2 = 2.155\text{E} + 19$$

$$\Sigma(x_i - \bar{x})^2 = 371337{,}4667$$

Maka:

$$S\widehat{\beta_1} = \sqrt{\frac{2.155\text{E} + 19}{30 - 2} / 371337{,}4667}$$

$$S\widehat{\beta_1} = 1439651$$

Statistik Uji:

$$t_{hitung} = \frac{7311897}{1439651}$$

$$t_{hitung} = 5{,}079$$

Nilai Kritis:

$$\alpha = 0{,}05$$

$t_{0.025,28} \approx 2{,}048$

```
Residuals:
       Min         1Q     Median         3Q        Max
-1.207e+09 -5.148e+08 -2.153e+08  3.928e+08  2.725e+09

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  859037027  239339148   3.589  0.00125 **
land_size_m2   7311897    1439651   5.079 2.23e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 877300000 on 28 degrees of freedom
Multiple R-squared:  0.4795,	Adjusted R-squared:  0.4609
F-statistic:  25.8 on 1 and 28 DF,  p-value: 2.235e-05
```

Kesimpulan:

Karena:

$$|5.079| > 2{,}048$$

Berdasarkan hasil perhitungan diperoleh nilai $t_{hitung} = 5.079$ dengan derajat bebas 28. Karena $t_{hitung} > t_{0.025,28}$ maka H_0 ditolak. Dengan demikian dapat disimpulkan bahwa luas tanah **berpengaruh signifikan** terhadap harga rumah.

Rumus Selang kepercayaan:

$$\widehat{\beta_1} \pm t_{\alpha/2,n-2} \times S\widehat{\beta_1}$$

Hitung:

$t_{0.025,28} \times S\widehat{\beta_1} = \ 2.948.993$

Batas Selang:

Batas Bawah:

$$7311897 - 2948993$$
$$= 4.362.904$$

Batas Atas:

$$7311897 + 2948993$$

$$= 10.260.890$$

Selang Kepercayaan:

$$(4.362.904; 10.260.890)$$

2. Uji hipotesis dan Selang kepercayan (1- α) bagi parameter regresi β_0

$H_0: \beta_0 = 0$
$H_1: \beta_0 \neq 0$

Hipotesis nol menyatakan bahwa harga rumah ketika luas tanah sama dengan 0 tidak signifikan dalam model regresi, sedangkan hipotesis alternatif menyatakan bahwa harga rumah ketika luas tanah sama dengan 0 memiliki nilai yang signifikan dalam model regresi.

Rumus Statistik Uji:

$$t_{\text{hitung}} = \frac{\widehat{\beta_0}}{S\widehat{\beta_0}}$$

Dengan:

$$S\widehat{\beta_0} = \sqrt{\frac{JKS}{n-2}\left(\frac{1}{n} + \frac{\bar{\mathrm{x}}^2}{\Sigma(x_i - \bar{x})^2}\right)}$$

$$JKS = \Sigma(y_i - \hat{y})^2 = 2.155\mathrm{E} + 19$$

$$\Sigma(x_i - \bar{x})^2 = 371337{,}4667$$

Maka:

$$S\widehat{\beta_0} = \sqrt{\frac{2.155\mathrm{E} + 19}{28}\left(\frac{1}{30} + \frac{15260{,}48}{371337{,}4667}\right)}$$

$$S\widehat{\beta_0} = 239339148$$

Statistik Uji:

$t_{hitung} = \frac{859037027}{239339148}$

$t_{hitung} = 3{,}589$

Nilai Kritis:

$\alpha = 0{,}05$

$t_{0.025,28} \approx 2{,}048$

```
esiduals:
      Min         1Q     Median         3Q        Max
-1.207e+09 -5.148e+08 -2.153e+08  3.928e+08  2.725e+09

oefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  859037027  239339148   3.589  0.00125 **
land_size_m2   7311897    1439651   5.079 2.23e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

esidual standard error: 877300000 on 28 degrees of freedom
Multiple R-squared:  0.4795,    Adjusted R-squared:  0.4609
F-statistic:  25.8 on 1 and 28 DF,  p-value: 2.235e-05
```

Kesimpulan:

Karena:

$$|3{,}589| > 2{,}048$$

Berdasarkan hasil perhitungan diperoleh nilai $t_{hitung} = 3{,}589$ dengan derajat bebas 28. Karena $t_{hitung} > t_{0.025,28}$ maka H_0 ditolak. Dengan demikian dapat disimpulkan bahwa harga rumah ketika luas tanah sama dengan 0 memiliki nilai yang signifikan dalam model regresi.

Rumus Selang kepercayaan:

$$\widehat{\beta_0} \pm t_{\alpha/2,n-2} \times S\widehat{\beta_0}$$

Hitung:

$t_{0.025,28} \times S\widehat{\beta_0} = 490.166.575$

Batas Selang:

Batas Bawah:

$$859.037.027 - 490.166.575$$
$$= 368.773.006$$

Batas Atas:

$$859.037.027 + 490.166.575$$

$$= 1.349.301.047$$

Selang Kepercayaan:

$$(368.773.006; 1.349.301.047)$$

```
                    2.5 %     97.5 %
(Intercept)   368773006 1349301047
land_size_m2    4362905   10260889
```

C. Tabel Sidik Ragam

Sumber Keragaman	db	JK	KT
Regresi	1	$\Sigma(\hat{y}_i - \bar{y})^2$	$\frac{JK_{\{Regresi)}}{1}$
Sisaan	n−2	$\Sigma(y_i - \hat{y}_i)^2$	$\frac{JK_{\text{sisaan}}}{n-2}$
Total	n−1	$\Sigma(y_i - \bar{y})^2$	

$$JKR = \Sigma(\hat{y}_i - \bar{y})^2 = 1.9853 \times 10^{19}$$

$$JKS = \Sigma(y_i - \hat{y}_i)^2 = 2.1550 \times 10^{19}$$

$$JKT = JKR + JKS = 4.1403 \times 10^{19}$$

$$KTR = \frac{JK_{\{Regresi)}}{1} = \frac{1.9853 \times 10^{19}}{1} = 1.9853 \times 10^{19}$$

$$KTG = \frac{JK_{\text{sisaan}}}{n-2} = \frac{2.1550 \times 10^{19}}{30-2} = 7.6963 \times 10^{17}$$

Sumber Keragaman	db	JK	KT
Regresi	1	1.9853×10^{19}	1.9853×10^{19}
Sisaan	28	2.1550×10^{19}	7.6963×10^{17}
Total	29	4.1403×10^{19}	

```
Analysis of Variance Table

Response: price_in_rp
             Df     Sum Sq    Mean Sq F value    Pr(>F)    
land_size_m2  1 1.9853e+19 1.9853e+19  25.796 2.235e-05 ***
Residuals    28 2.1550e+19 7.6963e+17                      
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

D. Koefisien Determinasi

Rumus Koefisien Determinasi:

$$R^2 = \frac{JK_{Regresi}}{JK_{Total}} = \frac{1.9853 \times 10^{19}}{4.1403 \times 10^{19}}$$

$$R^2 = 0{,}4795$$

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  859037027  239339148   3.589  0.00125 **
land_size_m2   7311897    1439651   5.079 2.23e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 877300000 on 28 degrees of freedom
Multiple R-squared:  0.4795,	Adjusted R-squared:  0.4609
F-statistic:  25.8 on 1 and 28 DF,  p-value: 2.235e-05
```

E. Selang Kepercayaan Individu dan Rataan

untuk variabel bebas x_0 saya akan mengambil luas tanah = 100 m², karena nilai ini berada di dalam rentang amatan

$$\widehat{y_0} = \widehat{\beta_0} + \widehat{\beta_1} x_0$$

$$\widehat{y_0} = 859037027 + (7311897)(100)$$

$$\widehat{y_0} = 1590226697$$

1. Selang Kepercayaan Rataan (Confidence Interval)

$$\widehat{y_0} \pm t_{\alpha/2,n-2} \cdot \sqrt{\text{KTS}\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\Sigma(x_i - \bar{x})^2}\right)}$$

$$t_{\alpha/2,n-2} \cdot \sqrt{\text{KTG}\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\Sigma(x_i - \bar{x})^2}\right)} = t_{0{,}025;28} \sqrt{7.6963 \times 10^{17}\left(\frac{1}{30} + \frac{553{,}817}{371337{,}5}\right)}$$

$$t_{\alpha/2,n-2} \cdot \sqrt{\text{KTG}\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\Sigma(x_i - \bar{x})^2}\right)} \approx 335.262.667$$

Batas Bawah:

$$1.590.226.697 - 335.262.667$$
$$\approx 1254873892$$

Batas Atas:

$$1.590.226.697 + 335.262.667$$
$$\approx 1925579503$$

Selang Kepercayaan Rataan:

$$(1254873892; 1925579503)$$

```{r}
predict(model,
        newdata=data.frame(land_size_m2=100),
        interval="confidence")
```

```
         fit        lwr        upr
1 1590226697 1254873892 1925579503
```

2. Selang Kepercayaan Individu (Prediction Interval)

$$\widehat{y_0} \pm t_{\alpha/2,n-2} \cdot \sqrt{\text{KTS}\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}$$

$$t_{\alpha/2,n-2} \cdot \sqrt{\text{KTS}\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\Sigma(x_i - \bar{x})^2}\right)}$$
$$= t_{0,025;28} \sqrt{7.6963 \times 10^{17}\left(1 + \frac{1}{30} + \frac{553,817}{371337,5}\right)}$$

$$t_{\alpha/2,n-2} \cdot \sqrt{\text{KTS}\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\Sigma(x_i - \bar{x})^2}\right)} \approx 18277 \times 10^9$$

Batas Bawah:

$$1.590.226.697 - 18277 \times 10^9$$
$$\approx -238697158$$

Batas Atas:

$$1.590.226.697 + 18277 \times 10^9$$
$$\approx 3418290553$$

Selang Kepercayaan Rataan:

$$(-238697158; 3418290553)$$

```
```{r}
predict(model,
        newdata=data.frame(land_size_m2=100),
        interval="prediction")
```

         fit   lwr   upr
1 1590226697 -237837158 3418290553
```